

P6-UC Uncooled Infrared Camera

Product Instruction

V1.0 May 2025



640 uncooled infrared camera

Directory

目录

Table of Contents	3
1. Description	4
2 Camera parameters	5
3 Pin Definition and Description	7
3.1 Pin diagram	7
3.2 Pin Description	7
4 serial port protocol	8
Serial port parameters:	8
Package Format:	8
4.1 Serial number read/write	8
4.1.1 Read Serial Number	8
Movement: -> upper computer	8
4.2 Version number read	9
4.3 baffle control	9
4.3.1 automatic correction	9
4.3.2 Manual calibration	9
4.3.3 Background correction	9
4.3.4 Set the automatic correction interval	10
4.3.5 The baffle is normally open	10
4.3.6 baffle normally closed	10
4.4 Save Parameters	10
4.5 Image Effect Adjustment	11
4.5.1 Brightness adjustment	11
4.5.2 Contrast adjustment	11
4.5.3 Enhanced regulation	11
4.5.4 denoising regulation	12
4.6 Zoom in	12
4.7 Mirror	12
4.7.1 Unmirror	12
4.7.2 Horizontal Mirroring	12
4.7.3 Vertical Mirror	13
4.7.4 diagonal mirror	13
4.8 pseudo color	13
4.8.1 white heat	13
4.8.2 black hot	14
4.8.3 Rainbow	14
4.8.4 green hot	14
4.8.5 red hot	14
4.8.6 iron red	15
4.9 Restore factory settings	15
5 Dimensional information	16

Table of Contents

7 Table 3- 1 Pin Description

Figure Catalog

7 Figure 3- 1 Pin diagram

1. Description

This infrared camera adopts uncooled vanadium oxide infrared focal plane detector, which is suitable for long-wave infrared (8-14 μ m) Radiation imaging, the camera is small and light, low power consumption, can be matched with different infrared lenses, support mainstream CVBS,USB video output interface, widely used in industrial acquisition, observation equipment, UAV and other fields of infrared thermal imaging image acquisition.

2 Camera parameters

Detector	Sensor Type	Vanadium oxide (VOx)	
	Image Size	640 x512	
	Pixel Size	12μm	
	response band	8~14μm	
	NETD (Noise Equivalent Temperature Difference)	≤40 mK	
	Frame Rate	50Hz	
Image Processing	Bad point correction	Support	
	Image detail enhancement	Support	
	Image noise reduction	Support	
	AGC mode	Histogram	
	Boot time	≤5S	
Function List	electronic amplification	x1-x4 0.1, Step	
	Brightness adjustment	Support	
	Contrast adjustment	Support	
	pseudo-color mode	White hot, black hot, rainbow, green hot, iron red	
	image calibration	Manual/automatic (with blank)/background	
Power	Power supply	4.5-18V	
	Power consumption	≤ 0.5W (default)	≤0.5W(USB UVC)
Interface	Video Interface	CVBS (default)/USB 2.0	
	Control Interface	UART 3.3V TTL	
	Output resolution	720x576(CVBS)	Native resolution (USB)
Environmental parameters	Working temperature	-40 °C ~ +60 °C	
	Storage temperature	-50 °C ~ +70 °C	
	Storage humidity	<90% RH	
	vibration shock	Vibration: sinusoidal vibration, 10 m/s, 3 axial Impact: half sine wave, 15g/11ms,3 axis 6	

Structural physics parameters	Weight	≤30g
	Dimensions (with lens)	21mm x 21mm x 37.3mm

3 Pin Definition and Description

3.1 Pin diagram



Fig. 3- 1 Pin Location

3.2 Pin Description

Table 3- 1 Pin Description

NO	Name	Type	Description
1	UART_RXD	IN	UART receive, 3.3V level
2	UART_TXD	OUT	UART transmit, 3.3V level
3	CVBS	OUT	Analog Video Output
4	GND	POWER	Negative pole of power supply
5	VIN	POWER	Power positive, range: 4.5~18V

Table 3- 1 Pin Description

NO	Name	Type	Description
1	VCC_	POWER	Power supply positive: 5V
2	GND_	POWER	Negative pole of power supply
3	USB_DP	1/O	USB2.0 signal line
4	USB_DM	1/O	USB2.0 signal line

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Upper computer: -> Movement

```
0x 55 0x 24 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

Movement: -> upper computer

```
0x 55 0x 24 0x 00 0x 01 0x 00 0x 00 0 x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00
```

The red 4 byte indicates the current version number, the first two bytes are the software version, and the third 4 byte is the hardware version. The green 1 byte is a check word.

Upper computer: -> Movement

```
0x 55 0x 26 0x 01 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

Movement: -> upper computer

```
0x 55 0x 26 0x 01 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

The red 1 byte indicates whether automatic correction is on, 0x 01 is on, 0x 00 is off. The green 1 byte is the check word

Upper computer: -> Movement

```
0x 55 0x 26 0x 02 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

Movement: -> upper computer

```
0x 55 0x 26 0x 02 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

The green 1 byte is the check word

Upper computer: -> Movement

```
0x 55 0x 26 0x 03 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

The green 1 byte is the check word

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Upper computer: -> Movement

```
0x 55 0x 2A 0x 04 0x 40 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

Movement: -> PC

```
0x 55 0x 2A 0x 04 0x 40 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

The red two bytes are denoising values, the little end indicates that the value range is 0-255, and the green 1 bytes are check words.

Upper computer: -> Movement

```
0x 55 0x 2B 0x 00 0x 10 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 x 00 0xAA
```

Movement: -> PC

```
0x 55 0x 2B 0x 00 0x 10 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 x 00 0xAA
```

The red 1 byte is the enlarged value, decimal means 1 times, 11 means 1.1 The maximum is 40 times 4 times. The green 1 byte is the check word

Upper computer: -> Movement

```
0x 55 0x 2C 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

Movement: -> upper computer

```
0x 55 0x 2C 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

The green 1 byte is the check word

Upper computer: -> Movement

```
0x 55 0x 2C 0x 01 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

The green 1 byte is the check word

Upper computer: -> Movement

```
0x 55 0x 2D 0x 05 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

Movement: -> PC

```
0x 55 0x2D 0x 05 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

The green 1 byte is the check word

4.9 Restore factory settings

PC -> Movement:

```
0x 55 0x2E 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

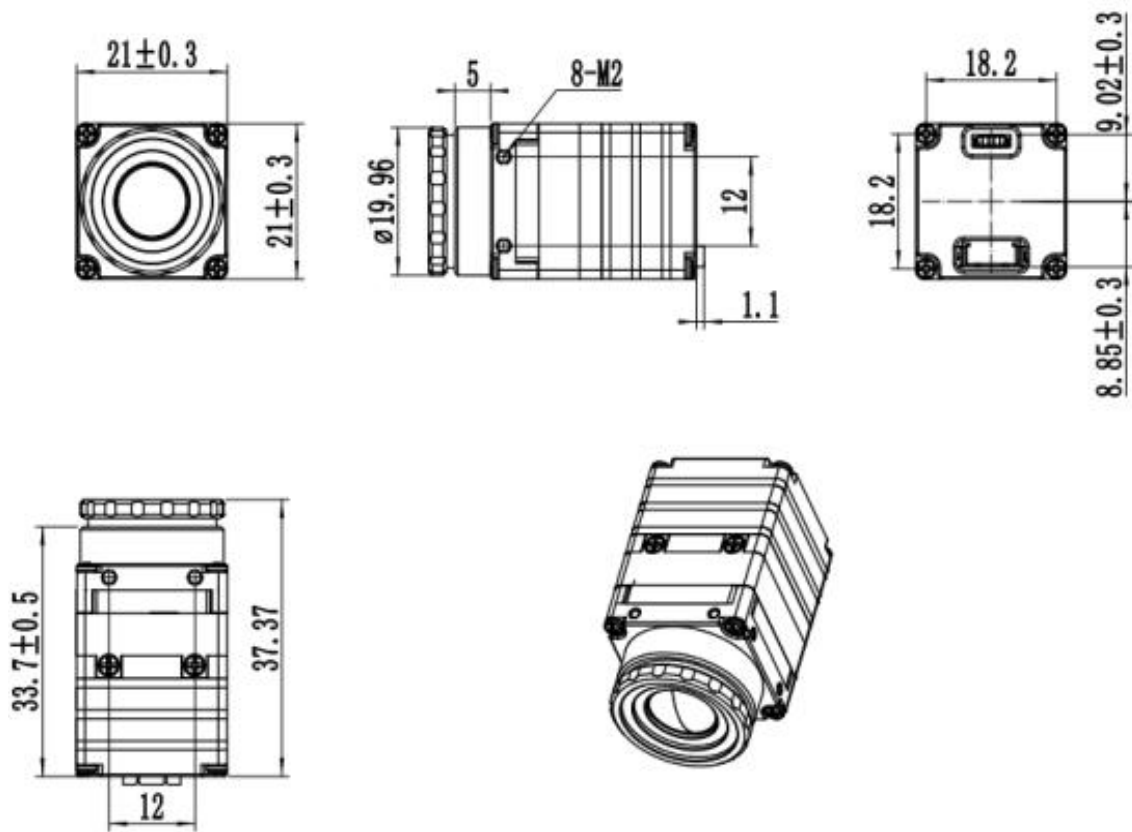
Movement -> upper computer:

```
0x 55 0x2E 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x 00 0x AA
```

The green 1 byte is the check word



5 Dimensional information



Revision History

Version	Date	Content
1.0	May 2025	First Edition Release

Document Description

Due to version upgrade or other reasons, the contents of this document will be updated from time to time. Unless otherwise agreed, the content of this document is only a guide to use, and all statements, information and recommendations in this document do not constitute any express or implied warranty.